$$u_n(x,t) = d_n \exp\left[\frac{-(n+\frac{1}{2})^2\pi^2 kt}{L^2}\right]\cos\left[\frac{(n+\frac{1}{2})\pi x}{L}\right], \quad n = 0,1,2,\ldots.$$

By the superposition principle, we have the more general solution

$$u(x,t) = \sum_{n=0}^{N} d_n \exp\left[\frac{-(n+\frac{1}{2})^2\pi^2 kt}{L^2}\right]\cos\left[\frac{(n+\frac{1}{2})\pi x}{L}\right] \tag{11}$$

which is the unique solution of problem (9), when

$$f(x) = u(x,0) = \sum_{n=0}^{N} d_n \cos\left[\frac{(n+\frac{1}{2})\pi x}{L}\right]. \tag{12}$$

Remark. All of the terms of (11) tend to zero as $t \to \infty$. The physical reason is that heat is allowed to pass through the uninsulated end at $x = L$. The first term ($n = 0$) of (11) decays at a rate proportional to $\exp(-\frac{1}{4}\pi^2 kt/L^2)$ which is slower than the corresponding rate $\exp(-\pi^2 kt/L^2)$ for the case where both ends are maintained at 0 (cf. Section 3.1). In Chapter 4, we will prove that any "reasonably nice" function $f(x)$ can be approximated arbitrarily closely by a sum as in (12). Also, there is a maximum principle that governs solutions of (9), so that the solution is not unduly perturbed by approximating the initial temperature distribution (cf. Problem 11). Typical graphs of the terms of (11) are shown in Figure 2. ∎

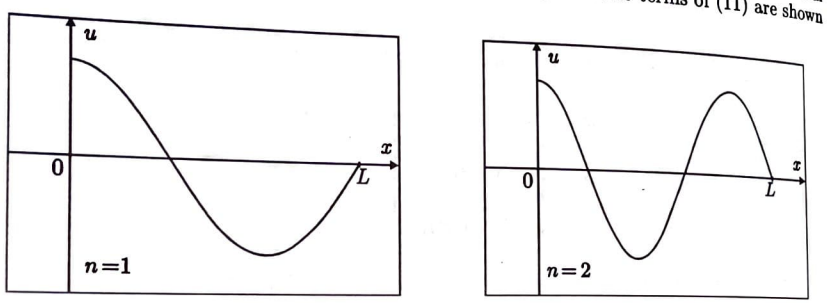

Figure 2

Inhomogeneous boundary conditions

The B.C. in Examples 1 and 2 [or in equation (1)] are homogeneous, whence we were able to use the superposition principle in forming more general solutions of the D.E. and B.C.. We now turn to the situation where the B.C. are not both homogeneous, but are still time-independent.

> The method of solution is to first find a *particular solution* of the D.E. and B.C., and then add (to that particular solution) the solution of a related problem with homogeneous B.C..

We have used this general idea before. The method is illustrated in the following example.

Example 3. For arbitrary real constants, a and b, and suitable $g(x)$, solve the problem

$$\begin{aligned} &\text{D.E.} \quad u_t = ku_{xx} \quad 0 \le x \le L, \quad t \ge 0 \\ &\text{B.C.} \quad u(0,t) = a \quad u(L,t) = b \\ &\text{I.C.} \quad u(x,0) = g(x). \end{aligned} \tag{13}$$

Solution. We first seek a particular solution $u_p(x,t)$ of the D.E. and B.C.. Since any particular solution will do, we may as well strive for simplicity. Indeed, a Case 3 product solution (cf. (13) of Section 3.1) $u_p(x,t) = cx + d$ will do, if c and d are chosen so that the B.C. are satisfied:

$$a = u_p(0,t) = c \cdot 0 + d = d$$
$$b = u_p(L,t) = cL + d = cL + a.$$

Thus, $d = a$ and $c = (b-a)/L$ and

$$u_p(x,t) = \frac{(b-a)x}{L} + a \tag{14}$$

solves both the D.E. and B.C.. Consider now the **related homogeneous problem** (i.e., with homogeneous D.E. and B.C.)

$$\begin{aligned} &\text{D.E.} \quad v_t = kv_{xx} \quad 0 \le x \le L, \quad t \ge 0 \\ &\text{B.C.} \quad v(0,t) = 0 \quad v(L,t) = 0 \\ &\text{I.C.} \quad v(x,0) = g(x) - u_p(x,0). \end{aligned} \tag{15}$$

If $g(x) - u_p(x,0)$ is of the form $\sum_{n=1}^{N} b_n \sin(\frac{n\pi x}{L})$, then we can solve this problem, obtaining

$$v(x,t) = \sum_{n=1}^{N} b_n e^{-(n\pi/L)^2 kt} \sin\left(\frac{n\pi x}{L}\right) \tag{16}$$

(cf. Proposition 1, Section 3.1). Now, set $u(x,t) = u_p(x,t) + v(x,t)$. We easily check that $u(x,t)$ solves problem (13). Indeed, $u(x,t)$ solves the D.E. by the superposition principle, and we have

$$\begin{aligned} &\text{B.C.} \begin{cases} u(0,t) = u_p(0,t) + v(0,t) = a + 0 = a \\ u(L,t) = u_p(L,t) + v(L,t) = b + 0 = b \end{cases} \\ &\text{I.C.} \quad u(x,0) = u_p(x,0) + v(x,0) = u_p(x,0) + g(x) - u_p(x,0) = g(x). \end{aligned} \tag{17}$$

Observe that *it is necessary to subtract* $u_p(x,0)$ from $g(x)$, when forming the I.C. of the related homogeneous problem (15). Otherwise the cancellation of $u_p(x,0)$ in (17) does not occur, and $u(x,0) = g(x)$ will not hold. ∎

Example 4. Solve

$$\begin{aligned} &\text{D.E.} \quad u_t = ku_{xx} \quad 0 \le x \le L, \quad t \ge 0 \\ &\text{B.C.} \quad u(0,t) = 0 \quad u(L,t) = L \\ &\text{I.C.} \quad u(x,0) = x + 3\sin\left(\frac{2\pi x}{L}\right). \end{aligned} \tag{18}$$

Solution. We can find a particular solution of the D.E. and B.C. of the form $u_p(x,t) = cx + d$. From the B.C., $0 = u(0,t) = c \cdot 0 + d = d$ and $L = u(L,t) = cL$. Thus, $d = 0$, $c = 1$ and $u_p(x,t) = x$. The related homogeneous problem is

$$\begin{aligned} &\text{D.E.} \quad v_t = kv_{xx} \quad 0 \le x \le L, \quad t \ge 0 \\ &\text{B.C.} \quad v(0,t) = 0 \quad v(L,t) = 0 \\ &\text{I.C.} \quad v(x,0) = x + 3\sin\left(\frac{2\pi x}{L}\right) - x = 3\sin\left(\frac{2\pi x}{L}\right), \end{aligned} \tag{19}$$

whose solution is $v(x,t) = 3e^{-4\pi^2 kt/L^2}\sin(\frac{2\pi x}{L})$. The solution of problem (19) is then

$$u(x,t) = u_p(x,t) + v(x,t) = x + 3e^{-4\pi^2 kt/L^2}\sin\left(\frac{2\pi x}{L}\right). \quad \blacksquare$$

Remark. It is probably best not to commit to memory formula (14). Instead the reader should consider a particular solution of the form $cx + d$, and find the constants, using the B.C.. The reason is that the formula only applies to the B.C. of (13). For other B.C., we obtain other particular solutions. For example,

$$\text{if } u_x(0,t) = a \text{ and } u(L,t) = b, \text{ then } u_p(x,t) = a(x - L) + b. \quad \blacksquare$$

Example 5. Solve

$$\begin{aligned} &\text{D.E.} \quad u_t = 2u_{xx} \quad 0 \le x \le 1, \quad t \ge 0 \\ &\text{B.C.} \quad u_x(0,t) = 1 \quad u(1,t) = -1 \\ &\text{I.C.} \quad u(x,0) = x + \cos^2\left(\frac{3\pi x}{4}\right) - \tfrac{5}{2}. \end{aligned}$$

Solution. We try a particular solution of the form $u_p(x,t) = cx + d$. The first B.C. yields $c = 1$, while $u_p(1,t) = 1 + d$ yields $d = -2$ by the second B.C.. Thus, $u_p(x,t) = x - 2$. The related homogeneous problem is

$$\begin{aligned} &\text{D.E.} \quad v_t = 2v_{xx} \quad 0 \le x \le 1, \quad t \ge 0 \\ &\text{B.C.} \quad v_x(0,t) = 0 \quad v(1,t) = 0 \\ &\text{I.C.} \quad v(x,0) = [x + \cos^2\left(\frac{3\pi x}{4}\right) - \tfrac{5}{2}] - (x - 2) = \tfrac{1}{2} + \tfrac{1}{2}\cos\left(\frac{3\pi x}{2}\right) - \tfrac{5}{2} + 2 = \tfrac{1}{2}\cos\left(\frac{3\pi x}{2}\right). \end{aligned}$$

From Example 2, we know that $v(x,t) = \frac{1}{2}e^{-9\pi^2 t/2}\cos(\frac{3\pi x}{2})$, and then

$$u(x,t) = x - 2 + \tfrac{1}{2}e^{-9\pi^2 t/2}\cos\left(\frac{3\pi x}{2}\right). \quad \blacksquare$$

Observe that in Examples 3, 4 and 5 the particular solution was time-independent, or in the usual terminology, **steady-state**.

> Any steady-state solution of the heat equation $u_t = ku_{xx}$ is of the form $cx + d$ (Why?).

In Examples 3, 4 and 5, the solutions $u(x,t) = u_p(x,t) + v(x,t)$ are sums of a steady-state particular solution of the D.E. and B.C. and the solution $v(x,t)$ (of the related homogeneous problem) which is **transient** in the sense that $v(x,t) \to 0$ as $t \to \infty$. Thus, in these examples,

$$u(x,t) = u_p(x,t) + v(x,t) \to u_p(x,t), \quad \text{as } t \to \infty$$

(i.e., the solution u approaches the steady-state solution as $t \to \infty$). However, for some types of B.C., there are no steady-state particular solutions, as the next example shows.

Example 6. For given real constants, a and b, and suitable $f(x)$, solve the problem

$$\begin{aligned} &\text{D.E.} \quad u_t = ku_{xx} \quad 0 \le x \le L, \quad t \ge 0 \\ &\text{B.C.} \quad u_x(0,t) = a \quad u_x(L,t) = b \\ &\text{I.C.} \quad u(x,0) = f(x). \end{aligned} \tag{20}$$

Solution. First note that the B.C. state that heat is being drained out of the end $x = 0$ at a rate $u_x(0,t) = a$ (cf. (2) of Section 3.1 with $KA = 1$) and heat is flowing into the end $x = L$ at a rate $u_x(L,t) = b$. If $b > a$, then the heat energy is being added to the rod at a constant rate. (If $b < a$, the rod loses heat at constant rate.) Thus, we cannot expect a steady-state solution of the D.E. and B.C., unless it happens that $a = b$. Indeed, putting $u_p(x,t) = cx + d$, the B.C. tell us that $c = a$ and $c = b$, which is impossible unless $a = b$. The next simplest form for a particular solution, that reflects the fact that the heat energy is changing at a constant rate, is

$$\boxed{u_p(x,t) = ct + h(x), \qquad (21)}$$

where c is a constant and $h(x)$ is a function of x. The constant c and the function $h(x)$ can be determined from the D.E. and B.C.. Indeed, a calculation shows that

$$c = (u_p)_t = k(u_p)_{xx} = kh''(x) \Rightarrow h''(x) = \frac{c}{k} \Rightarrow h(x) = \frac{c}{2k}x^2 + dx + e,$$

for constants d and e. The B.C. then yield

$$\left.\begin{aligned} a &= (u_p)_x(0,t) = h'(0) = d \\ b &= (u_p)_x(L,t) = h'(L) = \frac{cL}{k} + d \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} d &= a \\ c &= \frac{(b-a)k}{L}. \end{aligned}\right.$$

Thus, we arrive at the following particular solution (where we have set $e = 0$, for simplicity) of the D.E. and B.C. of (20):

$$u_p(x,t) = \frac{b-a}{L}kt + \frac{b-a}{2L}x^2 + ax = \frac{b-a}{L}\left(kt + \tfrac{1}{2}x^2\right) + ax. \tag{22}$$

The related homogeneous problem is

$$\text{D.E.} \quad v_t = kv_{xx} \qquad 0 \le x \le L, \quad t \ge 0$$

$$\text{B.C.} \quad v_x(0,t) = 0 \qquad v_x(L,t) = 0$$

$$\text{I.C.} \quad v(x,0) = f(x) - u_p(x,0) = f(x) - \left(\frac{b-a}{2L}x^2 + ax\right).$$

In the event that $f(x) - u_p(x,0)$ is of the form $\sum_{n=0}^{N} a_n \cos(\frac{n\pi x}{L})$, we have the solution

$$\begin{aligned} u(x,t) &= u_p(x,t) + v(x,t) \\ &= u_p(x,t) + \sum_{n=0}^{N} a_n e^{-(n\pi/L)^2 kt} \cos\left(\frac{n\pi x}{L}\right), \end{aligned} \tag{23}$$

where $u_p(x,t)$ is given by (22). ∎

Boundary conditions of the third kind

There are many other types of boundary conditions that can be imposed. For example, $u(0,t) - u_x(L,t) = 1$, $u_x(0,t) - [u(0,t)]^2 = 0$ or, more generally,

$$F(u(0,t), u(L,t), u_x(0,t), u_x(L,t)) = 0,$$

where the function F of four variables can be chosen at will. Even by restricting ourselves to those B.C. that have some physical relevance, there are more B.C. than we have time, space or endurance to handle. However, we will consider one additional type of B.C. that has great physical relevance, namely

$$\text{B.C.} \quad u_x(0,t) - cu(0,t) = 0 \qquad u_x(L,t) + c'u(L,t) = 0, \tag{24}$$

where c and c' are constants [usually positive in applications]. Note that as $c \to 0$, the first B.C. becomes $u_x(0,t) = 0$, while as $c \to \infty$, we obtain $u(0,t) = 0$ (and similarly

for the second B.C.). The B.C. (24) are known as **boundary conditions of the third kind**. The B.C. such as $u(0,t) = 0$ and $u_x(0,t) = 0$ are known as **B.C. of the first and second kinds** respectively; thus, these are limiting cases of the B.C. of the third kind. Essentially, the B.C. (24) arise because perfect insulation and perfect thermal contact (e.g., maintaining an end exactly at 0 by means of an external medium) are difficult to achieve in practice. To understand this more exactly, consider the picture (cf. Figure 3) near the end $x = 0$:

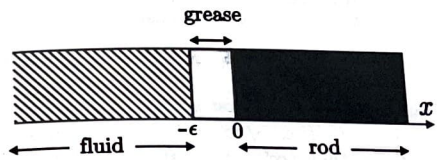

Figure 3

The region $x < -\epsilon$ is occupied by some fluid which is well circulated and maintained at temperature 0, while the small region $-\epsilon \le x \le 0$ is occupied by some intermediate substance (say grease, oxide or imperfect insulation). Assume that this substance is in perfect thermal contact with the rod and the fluid so that $u(x,t)$ is continuous at the junctions $x = -\epsilon$ and $x = 0$. In order that the flux of heat energy across $x = 0$ be the same when measured on either side of $x = 0$, the temperature gradient must suffer a jump so that

$$k_1 u_x(0^-,t) = ku_x(0^+,t), \tag{25}$$

where k_1 is the diffusivity constant for the intermediate substance (and k is that for the rod). Since the heat lost through $x = -\epsilon$ is assumed to be carried away immediately by circulating currents in the fluid, we have $u(-\epsilon,t) = 0$. If $u(0,t)$ is constant, then eventually a steady-state temperature distribution would result in $[-\epsilon, 0]$. Indeed, the transient term decreases as $\exp(-\pi^2 k_1 t/\epsilon^2)$. This decrease is especially rapid if ϵ is small, so that $k_1\pi^2/\epsilon^2 \gg 1$. If $u(0,t)$ is not constant, but varies rather slowly, then we still expect that the solution in the interval $[-\epsilon, 0]$ will be rather close to a linear (steady-state) distribution, i.e.,

$$u(x,t) \approx \frac{x+\epsilon}{\epsilon}u(0,t), \qquad -\epsilon < x \le 0,$$

when $k_1\pi^2/\epsilon^2 \gg 1$. Then $u_x(0^-,t) \approx \frac{1}{\epsilon}u(0,t)$, and (25) yields (to good approximation)

$$u_x(0^+,t) - \frac{1}{\epsilon}\left(\frac{k_1}{k}\right)u(0,t) = 0, \tag{26}$$

which is the first B.C. of (24), where $c = \frac{1}{\epsilon}(k_1/k)$. If $\frac{1}{\epsilon}(k_1/k)$ is small (say by virtue of k being large compared to k_1), then the end $x = 0$ of the rod behaves as if it were insulated. If $\frac{1}{\epsilon}(k_1/k)$ is large, then a close thermal contact exists between the rod and the fluid (i.e., $u(0,t) \approx 0$, where 0 is the temperature of the fluid). At the other end $x = L$, a similar analysis applies. Thus,

$$u_x(L^-,t) + \frac{1}{\epsilon}\left(\frac{k_1}{k}\right)u(L,t) = 0, \tag{27}$$

since the derivatives change sign. Of course, it could happen that k_1 and ϵ may be different at the end $x = L$. Thus, c' need not equal c in (24), but it is true that c and c' are positive in this physical context. For the B.C. (24), it is not always easy to find the values λ for which there is a Case 1 product solution (cf. (11) of Section 3.1). In the exercises, the reader is asked to verify that λ (where $\lambda > 0$) must satisfy

$$(cc' - \lambda^2)\sin(\lambda L) + (c + c')\lambda\cos(\lambda L) = 0. \tag{28}$$

Assume that c and c' are nonnegative. First note that $\lambda = \sqrt{cc'} > 0$ is a solution of (28), if and only if $\cos(\sqrt{cc'}\cdot L) = 0$ (i.e., $\sqrt{cc'} = (n + \frac{1}{2})\frac{\pi}{L}$, for some $n = 1, 2, 3, \ldots$). If a solution λ is not of this form, then $(cc' - \lambda^2)\cos(\lambda L) \neq 0$ and dividing both sides of (28) by this, we obtain

$$\tan(\lambda L) = \frac{(c + c')\lambda}{\lambda^2 - cc'}. \tag{29}$$

The (positive) solutions λ of this equation can be roughly determined from the points of intersections of the graphs of the functions of λ appearing on both sides of (29). If $\tan(\lambda L) = \infty$ (i.e., $\cos(\lambda L) = 0$) and $\lambda = \sqrt{cc'}$, then this exceptional solution must be included as well. Actually, this is simply the case where the graphs of the functions on either side of (29) approach a common vertical asymptote (i.e., they "intersect at ∞"). For example, if $c = c' = \pi/2$ and $L = 1$, then we have the graphs of $\tan(\lambda)$ and $\pi\lambda/(\lambda^2 - \frac{1}{4}\pi^2)$, shown in Figure 4.

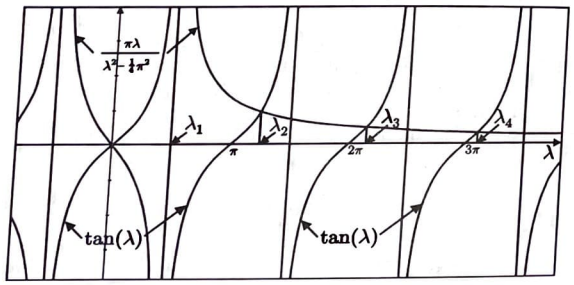

Figure 4

Here, $\lambda_1 = \pi/2$ corresponds to an intersection at infinity. The intersection at $\lambda = 0$ does not count, since we have assumed $\lambda > 0$. Numerical computation (cf. Problems 12 and 13 of Section 8.2) yields $\lambda_2 \approx 3.906$, $\lambda_3 \approx 6.741$, $\lambda_4 \approx 9.744$.

Remark. One can imagine physical situations where c and/or c' are negative. For instance, consider a rod immersed in a medium in which a heat-producing chemical (or

nuclear) reaction generates heat at a rate which is proportional to the local temperature. It is then possible that the portion of the medium which is close to a hot end of the rod may actually produce a heat influx into that end, at a rate which is positively proportional to the temperature at the end. In such a situation, there are likely to be Case 2 product solutions which grow exponentially with time, possibly leading to a melt-down or explosion (cf. Problem 9, parts (c) and (d)). ∎